

Regional/ethnic differences in ocular axial elongation and refractive error progression in myopic and non-myopic children

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Abstract

Aim: To determine the regional and ethnic differences in ocular axial elongation and refractive error progression in myopic and non-myopic children.

Methods: A retrospective analysis of 15 longitudinal clinical and population-based studies was conducted in the UK, Sweden, Australia (classified as European), China, and Vietnam (classified as East Asian) between 2005 and 2021. A total of 14,593 data points from 6208 participants aged 6–16 years with spherical equivalent from +6 to –6 D were analysed. Progression was annualised from longitudinal axial length and cycloplegic spherical equivalent (SE) refraction. Generalised estimating equation models including main effects and interactions were used for model building. Age and region-specific estimates for myopes and non-myopes and confidence intervals are reported.

Results: Factors affecting axial elongation and SE progression in children included being myopic, followed by age, region/ethnicity and sex. The magnitude of regional/ethnic differences was dependent on myopia and age. Axial elongation and SE progression were lower in European compared with East Asian children, but differences were reduced with increasing age and differences in axial elongation were larger in myopes than non-myopes. Age-specific regional/ethnic differences indicated that axial elongation for a 6-year-old East Asian myopic child was greater than a European child by 0.15 mm/year (0.58 vs. 0.43 mm/year) and by 0.09 mm/year (0.35 vs. 0.26 mm/year) for a 10-year-old myope. SE progression was lower in a 6-year-old European myope by 0.48 D/year and at 10 years of age by 0.34 D/year compared with an East Asian myope.

Conclusions: There are regional/ethnic differences in age-specific refractive and axial growth patterns in both myopic and non-myopic eyes, with more marked differences in younger East Asian children who demonstrated a higher axial growth and greater negative SE shift than their non-Asian peers. Regional/ethnic differences in progression reflect environmental and ethnic variations. Age and region/ethnicity-specific estimates could contribute as a reference for future comparisons.

KEYWORDS

axial length, children, ethnicity, myopia, progression, regional

INTRODUCTION

The evidence base around myopia and its clinical management has been advancing rapidly over the past few decades along with an increasing number of new interventions that have shown efficacy in reducing myopia progression.¹

Brennan et al.² argued that axial length growth rate should remain the focus of myopia management due to its precision, reliability, non-cycloplegic measurements and its usability across all myopia management options and monitoring for future myopic pathology.^{3–5} Following this, Wolffsohn et al. recommended using axial length as the primary outcome measure for monitoring myopia progression in combination with objective measurement of refractive error under cycloplegia.⁶ Monitoring spherical equivalent (SE) progression has the advantage of a functional outcome as it relates to tangible measures such as a change of spectacle prescription, thus providing greater patient engagement. In addition, optical biometers are not always available for axial length measurement in everyday clinical practice.

Recently, growth charts for axial length measurement have been developed in an effort to guide clinicians in myopia management and these have the added advantage of improving patient engagement.^{7–10} These charts, along with published data from several regions on myopia progression and axial elongation in children show the significant influence of age^{11–19} and ethnic differences, with children of Asian ethnicity having higher myopia progression¹³ and axial elongation.¹⁷ There is increasing debate as to whether the population differences observed are more genetically or environmentally driven. The evidence for a strong genetic pre-disposition to myopia due to racial differences alone appears to be lacking. Genome-wide association analysis shows that genetic factors for refractive error development were strongly correlated ($r > 0.78$) between ethnicities, indicating a lack of difference in genetic predisposition.²⁰ In contrast, there is increasing evidence implicating a stronger drive from environmental and behavioural factors.^{21–23}

Whilst there is evidence supporting the notion that there are regional and ethnic differences in axial length and myopia progression in myopic children, the extent of the differences is still being debated. Reported differences in annualised myopia progression between Asian and Caucasian children vary based on the source of data and type of analysis.^{12,13} Ethnic differences in axial elongation in myopes have been shown to be age-specific, with the differences decreasing with age,¹⁷ and differences in emmetropes are reported to be not significant.²⁴ However, most of these estimates are based on meta-analysis of summary statistics rather than raw data.

As the evidence for the efficacy of myopia control strategies continues to build, it is difficult to ethically justify randomising children into single vision control groups in clinical trials.²⁵ Hence, robust age and population

Key points

- Axial elongation and myopia progression vary by age and region/ethnicity, where East Asian eyes have higher age-specific progression compared with European eyes.
- Regional/ethnic differences are age-specific, where larger differences are observed in younger ages.
- Regional/ethnic differences in axial elongation are observed in both myopic and non-myopic eyes, but are greater in myopic eyes.

specific estimates of axial elongation and SE progression derived from multiple sources would help facilitate alternative approaches such as using a historical control group approach.²⁶

For improved generalisability of progression estimates, the availability of the raw data rather than summary findings from published studies^{13,17} is highly valuable. Furthermore, it is preferable to use a smaller proportion of clinical studies in comparison with population-based studies. Data derived from population-based studies detailing myopia progression is likely to be more representative of the general population rather than data taken predominantly from clinical studies due to the inherent selection bias in clinical studies towards progressing myopes. Moreover, population-based estimates from non-Asian regions report lower axial growth in Germany and the UK in comparison to China,^{8,10,16} and hence using data from regional sources to account for regional and ethnic differences in age-specific axial length growth and myopia progression is important.

Therefore, with an emphasis on population-based data, we analysed raw data retrospectively from Asian and European regions to determine regional/ethnic differences in axial elongation and SE progression in non-myopic and myopic children not undergoing myopia control intervention, as well as to improve the availability of normative progression data.

METHODS

Data for this retrospective analysis were obtained from clinical trials and population-based studies conducted in the UK, Sweden, Australia, Vietnam and China. Institutional human research ethics committee approval was obtained and the tenets of the Declaration of Helsinki for experimentation on humans were adhered to for all studies. Prior to data collection, written informed consent was obtained from the parent only or from the parent and participant.

Longitudinal data from China were derived from a dataset that was used in a previous publication,¹⁹ which

included six clinical trials conducted in Guangzhou and two population studies conducted in Shanghai. Data from Ho Chi Minh City, Vietnam included a clinical trial conducted between 2019 and 2021.^{27,28} All clinical trials investigated the efficacy of myopia control treatments in myopic children, who were followed for 6–36 months. Only data from the control single vision arm of clinical trials was used for this analysis.

Data from Australia comprised of two clinical trials conducted in Sydney, Australia between 2005–2008²⁹ and 2018–2019,³⁰ that investigated the efficacy of myopia control treatments in myopic children aged 8–16 years and followed for 6–36 months. Data from Europe were three population-based studies, one from Kalmar, Sweden³¹ and the other two from the UK (NICER1^{16,32} and NICER2³³). The three population studies in the European region included both non-myopic and myopic children who reported wearing single vision modality or no correction and had no prior myopia control treatment. The NICER1 was a longitudinal study conducted between 2006 and 2017 consisting of white children aged 6–7 and 12–13 years from schools in Northern Ireland, UK.³² Participating children were assessed at 3, 6 and 9 years from baseline. The NICER2 measured refractive error and biometry in a population-based sample of 6–7- and 12–13-year-old children in 2019.³³ Follow-up examinations were performed on a subgroup of participants after 12 months. The third population investigation was a refractive error study among Swedish schools that enrolled children aged 8–16 years and followed them annually for 2 years between 2019 and 2021.³¹

Details of study procedures from these studies have been described previously.^{19,31,32} All clinical trials measured refraction and ocular biometry in 6 monthly interval and population studies measured them annually, except NICER1 which conducted 3-yearly visits. Auto-refraction was used for refraction measurements under cycloplegia in all studies with Topcon auto-refractors (KR-8900; Topcon, [Topcon.co.jp](http://topcon.co.jp)) being utilised for two studies in China whilst all other studies used the Shin-Nippon open-field auto-refractor (SRW5000 or NVision-K 5001, rexam.co.jp/eye-care). Axial length measurements were performed using either an IOL Master (V3, V5, 500, 700, Carl Zeiss, zeiss.com) or Lenstar (LS900, Haag-Streit, mylenstar.com), both being partial coherence interferometry-based biometers.³⁴

Table 1 presents the study registration identifiers and study procedures for refraction and ocular biometry and Table 2 shows baseline demographic data of enrolled children within each region.

Statistical methods

Inclusion criteria were baseline age range of 6.0–16.9 years and baseline SE refractive error range within –6.00 and +6.00 Dioptres (D), which excluded high myopia and extreme values of the dataset's refractive error distribution. Myopia was defined as a SE refractive error of ≤ -0.50 D,

where SE was computed as sphere + cylinder/2. Myopia was defined for an eye, not a participant.

Data of Asians and non-Caucasians in the Australian sample were excluded from the analysis and combined with data from the European region and referred to as 'European' as all were of European ancestry and the majority were residing in Europe (98.7%). Similarly, data from Vietnam were combined with data from China and referred to as 'East Asian' as all were of Asian ancestry and the majority were residing in China, East Asia (98.6%).

Similar to a previous publication that used a portion of these data¹⁹ (Chinese study data), each 12-month period (baseline to 12, 12–24 and 24–36 months) was analysed to determine annualised progression after accounting for age at the start of each 12-month period. Progression was calculated as the difference from the 12-month visit to baseline, 24-month visit to 12-month visit and 36-month visit to 24-month visit. The exact follow-up time in years was computed as the difference between the start and end visit dates of each period. Progression was standardised to a 1-year period by multiplying progression from each period by the factor of 1/actual follow-up period in years and the resulting annualised SE progression and axial elongation were the outcomes used for this analysis.

Participants who were followed for >1 year contributed multiple time periods to the dataset. For example, a child observed at 12 and 24 months contributed two distinct annual time periods (baseline to 12, 12–24 months).

The NICER1 study follow-up was converted to three 36-month follow-up periods, namely, baseline to 3, 3–6 and 6–9 years, as all other studies had a maximum follow-up of 36 months. The age at the start of each period was considered as the age at baseline. Thereafter, the 36-month follow-up period was annualised by applying the divisor reported by Donovan et al.,¹³ which was 2.5 for Caucasians.

Prior to analysis, the following filters were applied to ensure that missing data, outliers, transcribing errors and exclusion criteria were appropriately handled prior to analysis: SE ≤ -6 D at the start of the follow-up period (0.6%), missing follow-up data for AL and SE (3.2%), outside the $3 \times$ standard deviation (SD) boundary limits applied to the linear relationship of SE progression versus axial elongation (2.5%),³⁵ SE progression and AL elongation outliers (0.30%). After outliers were removed, data points that had both AL shortening and SE regression beyond the instrument's repeatability were checked but none were identified. Finally, out of 15,611 participant-eye-periods, 14,593 (93.5%) were retained for analysis.

Across studies, different ages were sampled resulting in an unequal age distribution. Therefore, sample weights were applied to each age resulting in a uniform age distribution to obtain age-specific estimates. Moreover, studies spanned a wide time period ranging from 2005 to 2021, with unequal samples across this period. For example, the NICER1 study with a sample of 2332 participant-eye-periods commenced in 2006, whilst the NICER2 was in 2021 with a smaller sample of



TABLE 1 Details of study protocols.

Clinical trial/population study	Study registration identifiers	Number of participants	Cycloplegia protocol	Refraction	Ocular biometry
Clinical trials @ China ^a	ChiCTR-TRC-11001463 ⁴⁸ ChiCTR-OON-16008785 ⁴⁹ ChiCTR-IOR-17010432 ⁵⁰ ChiCTR-TRC-14004227 ⁵¹ ChiCTR-TRC-08000232 ^{52,53} ChiCTR-TRC-09000476 ⁵⁴	329	Two drops of tropicamide (1%) after corneal anaesthesia, 5 min apart; refraction after 30 mins	6 monthly using Shin Nippon SRW5000 auto refractor	6 monthly using Lenstar 900
Clinical trials @ Vietnam ^b	NCT04301323 ^{27,28}	68	Two drops of cyclopentolate (1%) after corneal anaesthesia, 5 min apart; refraction after 30 mins	6 monthly using Shin Nippon SRW5000 auto refractor	6 monthly using Lenstar 900
Population study 1 @ China	2015KY150 ⁵⁵	2521	Two drops of cyclopentolate (1%) after corneal anaesthesia, 5 min apart; refraction after 30 mins	Annually using Topcon KR-8900 auto refractor	Annually using IOL Master V5
Population study 2 @ China	NCT02980445 ⁵⁶	1851	Two drops of cyclopentolate (1%) after corneal anaesthesia, 5 min apart; refraction after 30 mins	Annually using Topcon KR-8900 auto refractor	Annually using IOL Master V5
Clinical trial @ Australia ^c	ACTRN12611001148965 ²⁹ ACTRN12618000242224 ³⁰	19	Two drops of cyclopentolate (1%) after corneal anaesthesia, 5 min apart; refraction after 30 mins	6 monthly using Shin Nippon SRW5000 auto refractor	6 monthly using Lenstar 900
Population study NICER1 @ Northern Ireland	REC/05/0121 ³²	721	One drop of cyclopentolate (1%) after corneal anaesthesia; refraction after 30 mins	3 yearly using Shin Nippon SRW5000 or NVision-K 5001 auto refractor	3 yearly using IOL Master V3
Population study NICER2 @ Northern Ireland	REC/18/0102 ³³	120	One drop of cyclopentolate (1%) after corneal anaesthesia; refraction after 30 mins	Annually using Shin Nippon SRW5000 or NVision-K 5001 auto refractor	Annually using IOL Master 700
Population study @ Sweden	DNR 2018/423-31 ³¹	124	Two drops of cyclopentolate (1%) after corneal anaesthesia, 10 min apart; refraction after 30 mins	Annually using Shin Nippon NVision-K 5001 auto refractor	Annually using IOL Master 500

^aRegistered in chictr.org.cn.^bRegistered in clinicaltrials.gov.^cRegistered in anzctr.org.au.

TABLE 2 Study sample and baseline characteristics by region.

Region	Study type	Participant-eyes			Ethnicity					Spherical equivalent at BL (D)	Myopic at BL (%)	
		At year 1	At year 2	At year 3	Total	Males (%)	2 myopic parents (%)	Asian (%)	Age at BL (years)			
East/Southeast Asian Region	Clinical trial	783	473	84	1340	50.4	31.8	100.0	10.9 ± 1.8 (6.3–16.0)	24.75 ± 0.82 (22.6–27.2)	−2.71 ± 1.01 (−5.8 to −0.4)	100.0
	Population study	5940	3365	866	10,171	53.8	19.9	100.0	8.2 ± 1.6 (6.0–16.9)	23.14 ± 0.93 (19.6–28.0)	0.41 ± 1.47 (−5.9 to 5.9)	22.5
European/Australian Region	Clinical trial	31	24	22	77	42.1	57.1	15.8	13.9 ± 1.9 (9.3–16.9)	24.40 ± 0.57 (23.5–26.1)	−2.52 ± 1.02 (−4.9 to −0.7)	100.0
	Population study	449	224	2332	3005	44.9	22.6	0.0	11.5 ± 3.1 (6.2–16.9)	23.13 ± 0.93 (19.7–27.5)	0.82 ± 1.39 (−5.8 to 5.9)	11.5
Total		7203	4086	3304	14,593	51.5%	21.1	76.9	9.1 ± 2.5 (6.0–16.9)	23.26 ± 1.02 (19.6–28.0)	0.27 ± 1.68 (−5.9 to 5.9)	25.7

Abbreviation: BL, baseline.

237 participant-eye-periods. To account for temporal changes (such as environment, urbanisation) over this period, sample weights were calculated using the formula $(17 - [2021 - \text{year of study commencement}])$. These temporal weights ensured that the most recent data had the highest weight in the analysis, that is, a study that commenced in 2021 would have a weight of 17 relative to a study that commenced in 2005 having a weight of 1. The two sets of weights, namely, the temporal weights and age weights, were multiplied to create sample weights and applied to the dataset prior to analysis.

Normality, was assessed using Q–Q plots. Axial elongation was natural log-transformed, which resulted in a normally distributed outcome and provided an exponential fit of the outcome variable. Natural log transformation was computed as $\ln(\text{axial elongation} + 1)$. Based on Q–Q plots, no transformation was required for SE progression.

Axial length elongation and SE progression were modelled using a generalised estimating equation (GEE) model with data from both eyes and multiple visits used for analysis. Multistage clustering effect of studies and repeat observations within a participant was accounted for using study and subject random intercepts and exchangeable correlation, which accounts for a homogenous correlation between repeats and a sandwich estimator of covariance. Baseline differences between studies within a region and between participants within a study were small, and such differences were accounted for by including the study and participants as random intercepts in the model, which also accounted for the correlation between the two eyes of participants. SE progression and natural log-transformed axial elongation were modelled using identity link function.

The initial model assessed whether the regional/ethnic differences (East Asian vs. European) interacted with the refractive error group (myopes vs. non-myopes). If there was a significant interaction, two separate GEE models were built for myopes and non-myopes. The non-myopic sample was categorised as either incident myopes (those who were non-myopic at baseline and became myopic by 12 months of follow-up) or persistent non-myopes (those who were non-myopic at baseline and at 12-month follow-up). The incident myopes were removed from the sample of non-myopes to obtain a more uniform sample of persistent non-myopes, labelled as non-myopes for this analysis. The factors for the GEE models included region/ethnicity (East Asian vs. European), age at baseline and sex. Main effects and two-way interaction terms were added to the model. The non-linear term of age was added if statistically significant. Modelling steps included forward entry of main effects and interactions, followed by backward elimination. Main and interaction effects were retained in the model if significant and improved the model fit. Final model equations were then used to estimate annualised age-specific axial elongation in millimetres and SE progression in dioptres. Estimated values of natural log-transformed elongation were back-transformed to obtain axial elongation. The width of the confidence interval was

based on the standard error of model estimates, whilst the width of the prediction interval was based on the model's error variance. The level of significance was set at 5% and statistical analysis was performed using IBM SPSS (v29) (ibm.com).

RESULTS

The dataset comprised 6208 unique participants aged between 6.0 and 16.9 years (9.1 ± 2.5 years) with 51.5% males and followed up to 36 months. Data from the East/Southeast Asian region was 76.8% of the sample. The proportion of children with myopia at baseline was 25.7%, refractive error (SE) ranged from -5.9 to $+5.9$ D (0.27 ± 1.68 D) and axial length ranged from 19.55 to 27.96 mm (23.26 ± 1.02 mm). At baseline, regional/ethnic differences (East Asian vs. European) were observed for age, sex and myopia prevalence. The dataset for East Asians had a higher proportion of males (54% vs. 45%, $p < 0.001$), and a younger age distribution (8.4 ± 1.8 vs. 11.5 ± 3.0 years, $p < 0.001$), though the age range was 6–16 years for both regions. The prevalence of myopia was also higher in the East Asians (30.9% vs. 12.4%, $p < 0.001$). Regional/ethnic differences were observed in ocular biometry at baseline. Axial length was higher in East Asians compared with

Europeans after accounting for baseline age, sex and SE (23.34 vs. 23.04 mm at baseline age of 9.4 years and SE of 0.27 D, $p < 0.01$). However, the difference in axial length by corneal curvature ratio (AL/CR) was not clinically significant (2.98 vs. 2.97 at baseline age of 9.8 years and SE of 0.17 D, $p < 0.01$). Due to these differences, age, sex and baseline SE were added in the model as confounders and retained even if they were not significant.

The analysis dataset comprised 14,593 participant-eye-period data points of which 25.0% were myopes at baseline, 7.5% were incident myopes and 67.5% were persistent non-myopes, with a higher proportion of incident myopes in East Asian children (9.1% vs. 1.8%).

Axial elongation and spherical equivalent progression

Scatter graphs of annualised axial elongation with SE progression for East Asian and European eyes are presented in [Figure 1](#) for all participants and [Figure 2](#) for myopes. As seen, East Asian eyes had a wider range of axial elongation and corresponding SE progression as compared with European eyes. The observations were similar for myopic eyes. The lines of best fit for East Asian and European eyes indicated that axial elongation and SE progression were

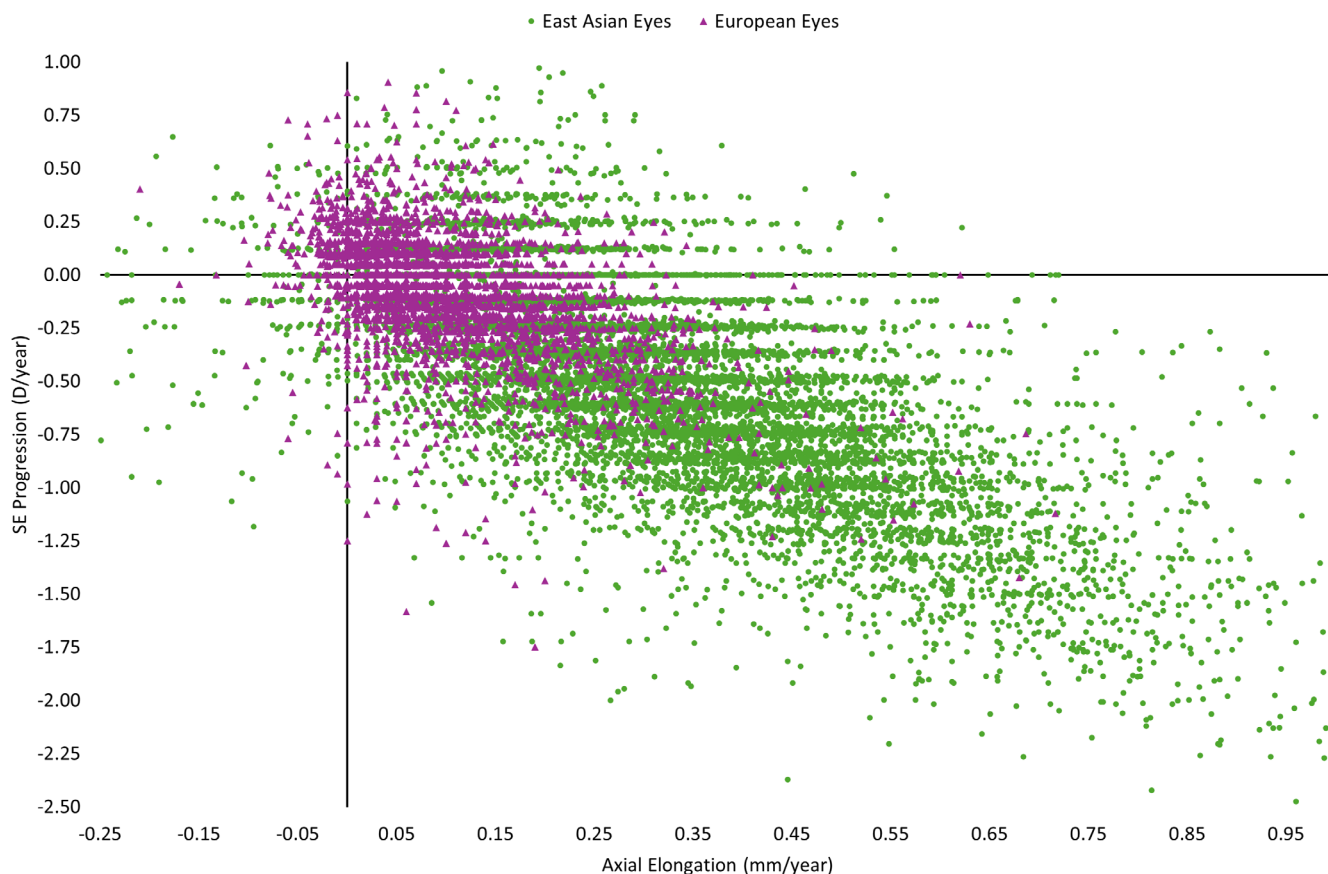


FIGURE 1 Scatter plot of annualised axial elongation (mm/year) with spherical equivalent (SE) progression (D/year) in East Asian and European eyes.

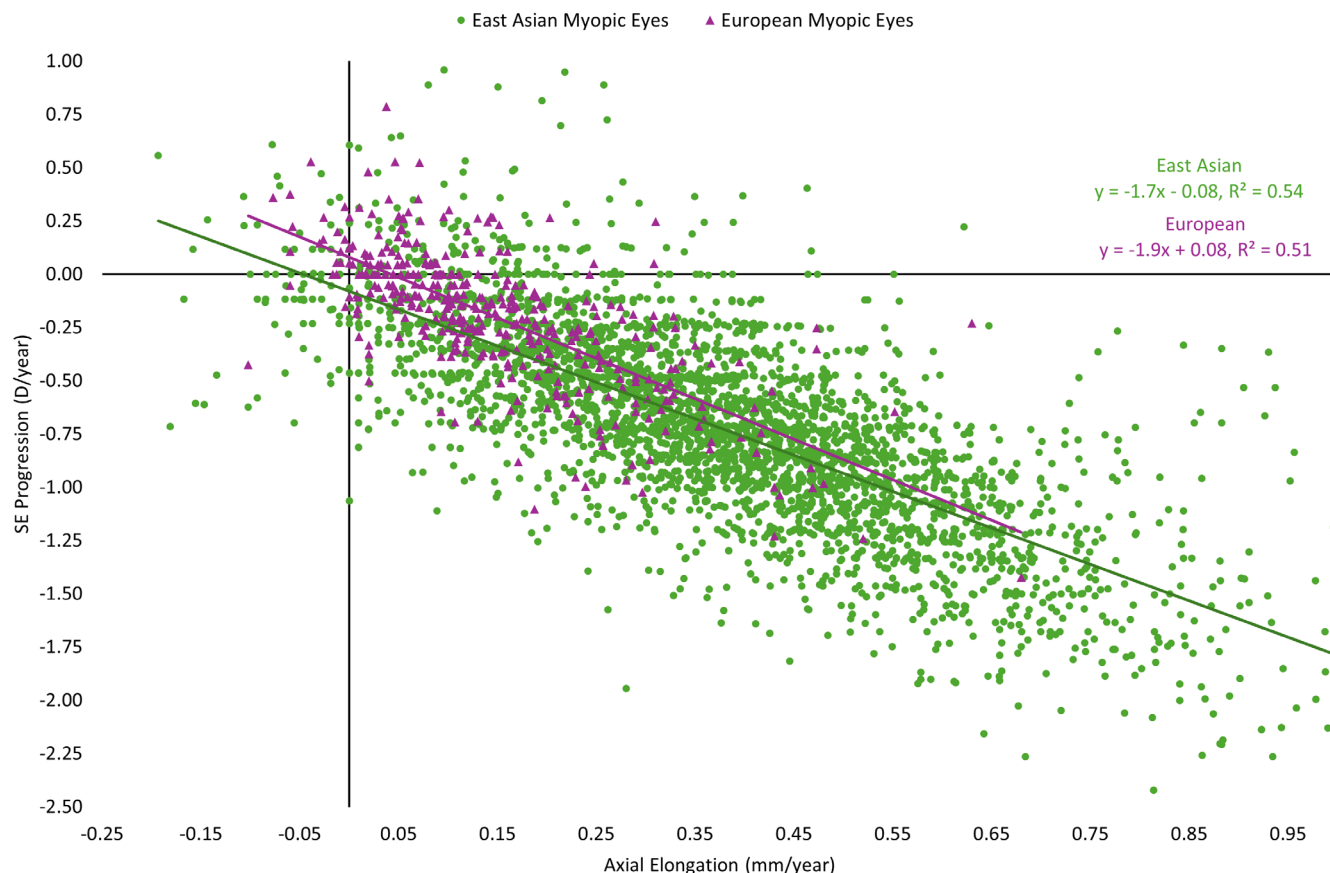


FIGURE 2 Scatter plot of annualised axial elongation (mm/year) with spherical equivalent (SE) progression (D/year) in East Asian and European myopic eyes.

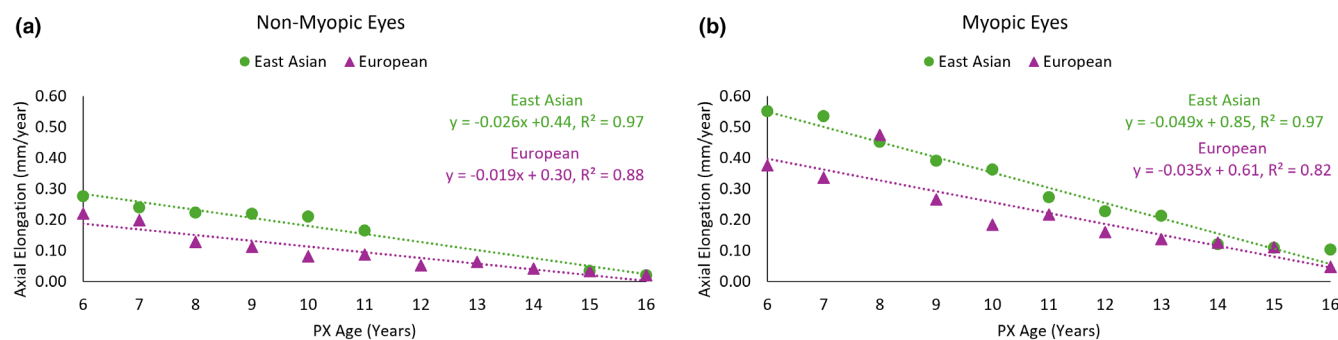


FIGURE 3 Observed mean axial elongation (mm/year) in East Asian and European non-myopic (a) and myopic (b) eyes by baseline age. Symbols denote observed weighted means. Dotted lines represent linear line of best fit. PX, participant.

linearly related with similar slopes for both groups (East Asian eyes slope = -1.7 D/mm , $r^2 = 0.54$ and European eyes slope = -1.9 D/mm , $r^2 = 0.51$). However, at low levels of axial elongation, East Asian eyes demonstrated higher SE progression than European eyes.

Axial elongation model

Observed annualised axial elongation summarised by baseline age is presented in Figure 3 for non-myopes and myopes by regional/ethnic groups. Axial elongation

significantly reduced with increasing age for all groups (myopes versus non-myopes and between regions). Based on the line of best fit, a larger regional/ethnic difference was observed in the younger ages and this reduced significantly with increasing age. The age-specific slopes indicated a steeper age gradient in East Asian myopes than European myopes and nearly two times higher slope in myopes compared with non-myopes.

An initial GEE modelling of the entire dataset indicated axial elongation was significantly higher in myopes compared with non-myopes ($p < 0.001$); that also interacted with the region ($p < 0.001$). This required that

the regional/ethnic differences in axial elongation were estimated separately for myopes and non-myopes, thus two separate GEE models were developed for myopic and non-myopic eyes. In both models, region/ethnicity, age at baseline, sex and SE at baseline were significantly associated with axial elongation. East Asian eyes had a higher eye growth rate compared with European eyes, and regional/ethnic differences were greater in myopes compared with non-myopes. Although axial elongation decreased with increasing age in both myopes and non-myopes, the rate of decrease was higher in myopes than non-myopes by a factor of two (beta coefficient for age = -0.032 mm/year for myopes and -0.016 mm/year for non-myopes). The relationship of age with axial elongation interacted with region, indicating that axial elongation had a stronger association with age in East Asian eyes. This interaction was similar in both myopes and non-myopes.

Although sex was a significant factor for both myopic and non-myopic children, the effect was small. In myopes, males had greater axial elongation than females by 0.004 mm/year, but in non-myopes, axial elongation was lower in males compared with females by 0.003 mm/year.

Baseline SE was a significant factor in both myopic and non-myopic models, where axial elongation slightly increased with more myopic SE, but the relationship was stronger in myopes (beta coefficient for SE = -0.007 mm/D for myopes and -0.002 mm/D for non-myopes).

The complete model equations for non-myopic and myopic eyes are presented in Table 3. The coefficients are the linear regression coefficients for natural log-transformed axial elongation (y), and can be used to derive estimates of axial elongation for specific factors using the equation axial elongation = Exponent (y) $- 1$. The coefficient of determination (R^2) between observed and predicted was 0.42 and 0.50 for non-myopic and myopic models, respectively. The model standard error to compute prediction intervals was 0.08 and 0.11 mm for non-myopic and myopic models, respectively, whilst the highest age-specific standard error to compute confidence intervals around the mean was 0.011 and 0.018 mm/year for non-myopic and myopic models, respectively. Annualised age-specific axial elongation estimates were derived from the model equations, assuming an even sex distribution (male: female—50% each). The estimates were derived for baseline SE of 0 to $+2$ D for

TABLE 3 Generalised estimating equation (GEE) model equation of factors associated with natural log-transformed axial elongation in non-myopic and myopic eyes.

Refractive error group	Factors	Beta coefficient ^a	Standard error	95% confidence interval		p-Value
				Lower	Upper	
Non-myopes (at baseline and 12 months)	Intercept	0.263	0.001	0.262	0.265	<0.001
	East Asian	0.136	0.001	0.134	0.139	<0.001
	European	0.000				
	Males	-0.003	0.0003	-0.004	-0.003	<0.001
	Females	0.000				
	Age at baseline (years)	-0.016	0.0001	-0.016	-0.016	<0.001
	East Asian: age at baseline (years)	-0.007	0.0001	-0.007	-0.007	<0.001
	European: age at baseline (years)	0.000				
Myopes at baseline	SE at baseline (D)	-0.002	0.0002	-0.002	-0.001	<0.001
	Intercept	0.537	0.003	0.531	0.542	<0.001
	East Asian	0.147	0.004	0.141	0.154	<0.001
	European	0.000				
	Males	0.004	0.001	0.002	0.005	<0.001
	Females	0.000				
	Age at baseline (years)	-0.032	0.0002	-0.032	-0.031	<0.001
	East Asian: age at baseline (years)	-0.007	0.0003	-0.008	-0.007	<0.001
	European: age at baseline (years)	0.000				
	SE at baseline (D)	-0.007	0.0003	-0.008	-0.006	<0.001

Abbreviations: ; SE, spherical equivalent.

^aLinear regression coefficients to predict natural log of axial elongation.

non-myopic eyes and -1 to -3 D for myopic eyes as baseline SE was a significant factor in the model, though contributing to a small effect. These refractive errors represented the largest proportion (91.5%) of refractive errors in the sample. Estimates of annual age-specific axial elongation and 99% confidence interval (99% was used instead of 95% due to small standard errors) are provided in Table 4 and graphically presented in Figure 4.

Model-based estimates of axial elongation indicated regional/ethnic differences were greater in the younger ages and in myopes. In non-myopes, axial elongation at 6 years of age was 0.30 and 0.18 mm/year for East Asians and Europeans, and at 16 years reduced to 0.03 mm/year for East Asians and 0.01 mm/year for Europeans, respectively. Regional/ethnic differences in axial elongation were age-specific, which ranged from 0.12 mm/year at 6 years of age to 0.02 mm/year at 16 years and 0.06 mm/year (40% reduced elongation in Europeans) at the median age of 11 years. Similarly, in myopes axial elongation peaked at 0.58 and 0.43 mm/year at 6 years of age and reduced to 0.07 and 0.04 mm/year at 16 years for East Asians and Europeans, respectively. Regional/ethnic differences in axial elongation among myopes were also age-specific, which ranged from 0.16 mm/year at 6 years of age to 0.03 mm/year at 16 years of age and 0.08 mm/year (28% reduced elongation in Europeans) at the median age of 11 years. Based on the estimates and associated confidence intervals, regional/ethnic differences were negligible at 14 years and older for both myopic and non-myopic children.

Spherical equivalent (SE) progression

A similar analysis was performed for SE progression. Figure 5 presents the observed annual SE progression by baseline age and region/ethnicity for non-myopic and myopic eyes. The line of best fit of age-specific SE progression indicated that SE progression was higher in East Asian myopic eyes and the regional/ethnic difference was larger in younger ages. In non-myopic eyes the regional/ethnic difference was small.

Like the axial length model, a significant interaction of region/ethnicity and refractive error groups ($p < 0.001$) indicated that separate models are required for non-myopes and myopes. Model factors, namely, region/ethnicity, age at baseline, sex and SE at baseline were significantly associated with SE progression, with East Asians showing a higher progression than Europeans for myopes and non-myopes ($p < 0.01$). However, the r^2 -value for the non-myopes model was < 0.1 . Due to this low value, modelling for non-myopes was not pursued further.

The progression model for myopes indicated a significantly greater progression in East Asians compared with Europeans, with the rate of progression decreasing with increasing age. Baseline age was factored with a linear

and quadratic term indicating that the rate of progression was higher in the younger age groups and tapered off for ages closer to 16 years. The association of age also interacted with region/ethnicity, indicating that the rate of SE progression with age was greater in East Asians than Europeans. Females had a marginally higher progression rate than males. Baseline SE added as a confounder was significant in the myopic progression model, where SE progression increased by 0.042 D/year with every 1 D increase in myopic SE at baseline, but reduced to clinically non-significant levels (0.004 D/year) in the East Asian sample.

The SE progression model equation for myopic eyes is presented in Table 5. The coefficients are the linear regression coefficients for SE progression (D/year). The coefficient of determination (R^2) between observed and predicted was 0.32 and the model standard error to compute prediction intervals was 0.40 D, whilst the highest age-specific standard error to compute confidence intervals around the mean was 0.07 D/year. Annualised age-specific progression estimates were derived from the model equations, assuming an even sex distribution (male: female—50% each) and for baseline SE between -1.0 and -3.0 D for myopic eyes. These estimates with 99% confidence intervals are provided in Table 6 and graphically presented in Figure 6.

The SE progression estimates indicated that regional/ethnic differences were greater in younger ages. Estimates indicated that SE progression was highest at 6 years of age (-1.08 D/year for East Asians and -0.60 D/year for Europeans with baseline SE of -1.0 D) and reduced to -0.17 D/year for East Asians and to non-significant levels for Europeans. The regional/ethnic differences in SE progression were age-specific, with a larger dioptric difference at 6 years of age (0.49 D), reducing to 0.12 D at 16 years. At the median age of 11 years, SE progression in myopes was lower in Europeans by 0.30 D/year (54% lower), whilst regional/ethnic differences appear to be significantly reduced at 13 years and older, based on estimate means and confidence intervals.

DISCUSSION

This retrospective analysis using a large longitudinal dataset of myopic and non-myopic eyes from population and clinical studies demonstrated regional and ethnic differences in age-specific axial elongation and SE progression. Particularly, children of Asian descent from China and Vietnam had higher age-specific myopia progression and axial elongation compared with children of European descent from the UK, Sweden and Australia. These regional/ethnic differences were age-specific, where larger differences were observed in younger ages. Moreover, the regional/ethnic differences in axial elongation were observed in both myopic and non-myopic eyes.

In classifying myopia progression, region and ethnicity may have significant overlap. Ethnicity is a term that

TABLE 4 Generalised estimating equation (GEE) model-based estimates of annualised axial elongation (mm/year) with 99% confidence limits in parentheses by baseline age and refractive error in non-myopic and myopic East Asian and European eyes.

Refractive error group	Age at baseline (yrs)	0D @ baseline		1D @ baseline		2D @ baseline	
		East Asian	European	East Asian	European	East Asian	European
Non-myopes (at baseline and 12 months)	6	0.30 (0.27–0.32)	0.18 (0.16–0.20)	0.30 (0.27–0.32)	0.18 (0.16–0.20)	0.29 (0.27–0.32)	0.18 (0.16–0.20)
	7	0.27 (0.24–0.29)	0.16 (0.14–0.18)	0.27 (0.24–0.29)	0.16 (0.14–0.18)	0.26 (0.24–0.29)	0.16 (0.14–0.18)
	8	0.24 (0.21–0.26)	0.15 (0.13–0.17)	0.24 (0.21–0.26)	0.14 (0.12–0.16)	0.23 (0.21–0.26)	0.14 (0.12–0.16)
	9	0.21 (0.19–0.24)	0.13 (0.11–0.15)	0.21 (0.18–0.23)	0.13 (0.11–0.15)	0.21 (0.18–0.23)	0.12 (0.10–0.14)
	10	0.18 (0.16–0.21)	0.11 (0.09–0.13)	0.18 (0.16–0.21)	0.11 (0.09–0.13)	0.18 (0.15–0.20)	0.11 (0.09–0.13)
	11	0.16 (0.13–0.18)	0.09 (0.07–0.11)	0.15 (0.13–0.18)	0.09 (0.07–0.11)	0.15 (0.13–0.18)	0.09 (0.07–0.11)
	12	0.13 (0.10–0.15)	0.08 (0.06–0.10)	0.13 (0.10–0.15)	0.07 (0.05–0.09)	0.13 (0.10–0.15)	0.07 (0.05–0.09)
	13	0.10 (0.08–0.13)	0.06 (0.04–0.08)	0.10 (0.08–0.13)	0.06 (0.04–0.08)	0.10 (0.08–0.13)	0.06 (0.04–0.08)
	14	0.08 (0.05–0.10)	0.04 (0.02–0.06)	0.08 (0.05–0.10)	0.04 (0.02–0.06)	0.08 (0.05–0.10)	0.04 (0.02–0.06)
	15	0.05 (0.03–0.08)	0.03 (0.01–0.05)	0.05 (0.03–0.08)	0.03 (0.01–0.05)	0.05 (0.03–0.08)	0.02 (0.00–0.04)
	16	0.03 (0.01–0.06)	0.01 (–0.01–0.03)	0.03 (0.00–0.05)	0.01 (–0.01–0.03)	0.03 (0.00–0.05)	0.01 (–0.01–0.03)
Refractive error group	Age at baseline (yrs)	–1D @ Baseline		–2D @ Baseline		–3D @ Baseline	
		East Asian	European	East Asian	European	East Asian	European
Myopes at baseline	6	0.58 (0.56–0.60)	0.43 (0.39–0.47)	0.59 (0.57–0.61)	0.44 (0.40–0.48)	0.60 (0.58–0.62)	0.45 (0.41–0.49)
	7	0.52 (0.50–0.54)	0.38 (0.34–0.42)	0.53 (0.51–0.55)	0.39 (0.35–0.43)	0.54 (0.52–0.56)	0.40 (0.36–0.44)
	8	0.46 (0.44–0.48)	0.34 (0.30–0.38)	0.47 (0.45–0.49)	0.35 (0.31–0.39)	0.48 (0.46–0.50)	0.36 (0.32–0.40)
	9	0.41 (0.39–0.43)	0.30 (0.26–0.34)	0.42 (0.40–0.44)	0.31 (0.27–0.35)	0.43 (0.41–0.45)	0.31 (0.27–0.35)
	10	0.35 (0.33–0.37)	0.26 (0.22–0.30)	0.36 (0.34–0.38)	0.26 (0.22–0.30)	0.37 (0.35–0.39)	0.27 (0.23–0.31)
	11	0.30 (0.28–0.32)	0.22 (0.18–0.26)	0.31 (0.29–0.33)	0.22 (0.18–0.26)	0.32 (0.30–0.34)	0.23 (0.19–0.27)
	12	0.25 (0.23–0.27)	0.18 (0.14–0.22)	0.26 (0.24–0.28)	0.19 (0.15–0.23)	0.27 (0.25–0.29)	0.19 (0.15–0.23)
	13	0.20 (0.18–0.22)	0.14 (0.10–0.18)	0.21 (0.19–0.23)	0.15 (0.11–0.19)	0.22 (0.20–0.24)	0.16 (0.12–0.20)
	14	0.16 (0.14–0.18)	0.11 (0.07–0.15)	0.16 (0.14–0.18)	0.11 (0.07–0.15)	0.17 (0.15–0.19)	0.12 (0.08–0.16)
	15	0.11 (0.09–0.13)	0.07 (0.03–0.11)	0.12 (0.10–0.14)	0.08 (0.04–0.12)	0.13 (0.11–0.15)	0.09 (0.05–0.13)
	16	0.07 (0.05–0.09)	0.04 (0.00–0.08)	0.08 (0.06–0.10)	0.04 (0.00–0.08)	0.08 (0.06–0.10)	0.05 (0.01–0.09)

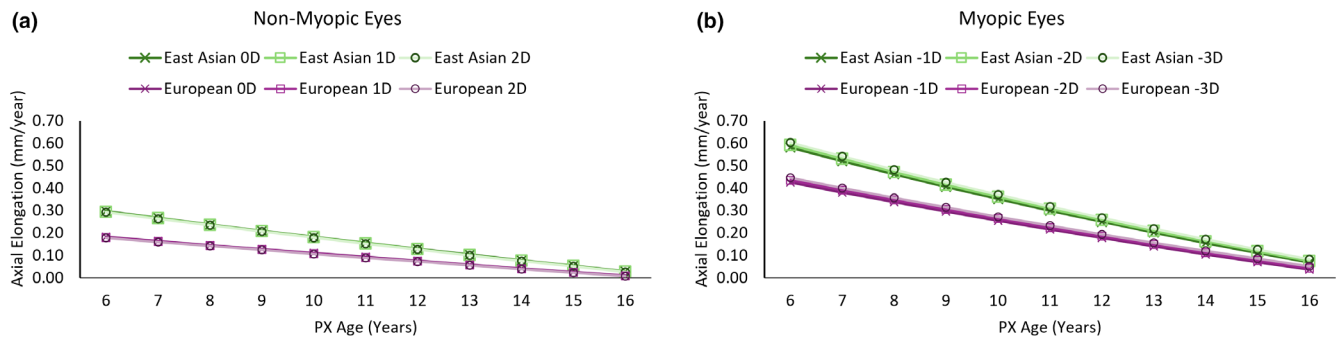


FIGURE 4 Annualised axial elongation (mm/year) estimated by generalised estimating equation (GEE) model by age at baseline in East Asian and European non-myopic (a) and myopic (b) eyes. PX, participant.

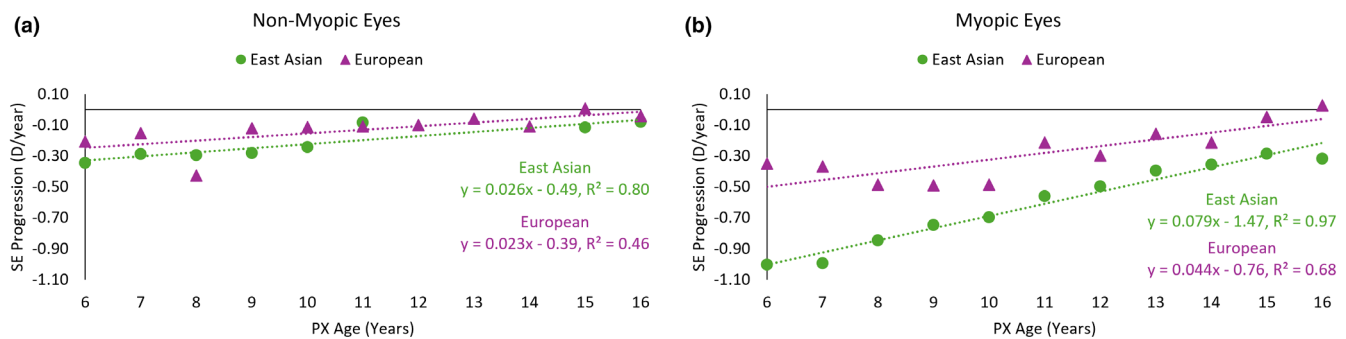


FIGURE 5 Observed mean spherical equivalent (SE) progression (D/year) in East Asian and European non-myopic (a) and myopic (b) eyes by baseline age. Symbols denote observed weighted means. Dotted lines represent linear line of best fit. PX, participant.

TABLE 5 Generalised estimating equation (GEE) model equation of factors associated with spherical equivalent progression in myopic eyes.

Refractive error group	Factors	Beta coefficient ^a	Standard error	95% confidence interval		p-Value
				Lower	Upper	
Myopes at baseline	Intercept	-1.119	0.024	-1.166	-1.072	<0.001
	East Asian	-0.702	0.011	-0.724	-0.681	<0.001
	European	0.000				
	Males	0.014	0.003	0.008	0.019	<0.001
	Females	0.000				
	Age at baseline (years)	0.107	0.004	0.099	0.115	<0.001
	Age at baseline squared	-0.002	0.0002	-0.003	-0.002	<0.001
	East Asian: age at baseline (years)	0.036	0.001	0.034	0.038	<0.001
	European: age at baseline (years)	0.000				
	SE at baseline (D)	0.042	0.001	0.039	0.044	<0.001

^aLinear regression coefficients to predict spherical equivalent (SE) progression.

suggests a combination of genetic, environmental, and socio-cultural factors,¹⁴ and has been implicated to be a significant factor associated with myopia progression and axial elongation within a common region. In an ethnically diverse population from the USA (COMET study), African-American children had the least axial elongation and myopia progression whereas Asian-American children had the highest, but were not significantly

different from Hispanic and white children.¹¹ Similarly, the CLEERE population-based study in the USA showed that on average, myopia progression in a 7-year-old was higher in Asian-American children compared with African-American children by 0.88 D and from Hispanics by 0.47 D, but higher than white children by only 0.19 D.¹² In an Australian urban population, the annual change of refraction in children of East Asian ethnicity was

TABLE 6 Generalised estimating equation (GEE) model-based estimates of annualised spherical equivalent progression (D/year) with 99% confidence limits in parentheses by baseline age and refractive error in myopic East Asian and European eyes.

Refractive error group	Age at baseline (years)	-1 D @ baseline		-2 D @ baseline		-3 D @ baseline	
		East Asian	European	East Asian	European	East Asian	European
Myopes at baseline	6	-1.08 (-1.18 to -0.98)	-0.60 (-0.76 to -0.43)	-1.12 (-1.22 to -1.02)	-0.64 (-0.81 to -0.47)	-1.16 (-1.26 to -1.06)	-0.68 (-0.85 to -0.51)
	7	-0.97 (-1.07 to -0.87)	-0.52 (-0.69 to -0.35)	-1.01 (-1.11 to -0.91)	-0.56 (-0.73 to -0.39)	-1.05 (-1.15 to -0.95)	-0.60 (-0.77 to -0.44)
	8	-0.86 (-0.96 to -0.76)	-0.45 (-0.62 to -0.28)	-0.90 (-1.00 to -0.80)	-0.49 (-0.66 to -0.32)	-0.94 (-1.04 to -0.84)	-0.53 (-0.70 to -0.36)
	9	-0.76 (-0.86 to -0.66)	-0.38 (-0.55 to -0.21)	-0.80 (-0.90 to -0.70)	-0.42 (-0.59 to -0.26)	-0.84 (-0.94 to -0.74)	-0.47 (-0.63 to -0.30)
	10	-0.66 (-0.76 to -0.56)	-0.32 (-0.49 to -0.15)	-0.70 (-0.80 to -0.60)	-0.36 (-0.53 to -0.19)	-0.74 (-0.84 to -0.64)	-0.40 (-0.57 to -0.24)
	11	-0.57 (-0.67 to -0.47)	-0.26 (-0.43 to -0.10)	-0.61 (-0.71 to -0.51)	-0.30 (-0.47 to -0.14)	-0.65 (-0.75 to -0.55)	-0.35 (-0.51 to -0.18)
	12	-0.48 (-0.58 to -0.38)	-0.21 (-0.38 to -0.04)	-0.52 (-0.62 to -0.42)	-0.25 (-0.42 to -0.08)	-0.56 (-0.66 to -0.46)	-0.29 (-0.46 to -0.13)
	13	-0.39 (-0.49 to -0.29)	-0.16 (-0.33 to 0.01)	-0.44 (-0.54 to -0.34)	-0.20 (-0.37 to -0.04)	-0.48 (-0.58 to -0.38)	-0.25 (-0.41 to -0.08)
	14	-0.32 (-0.42 to -0.22)	-0.12 (-0.29 to 0.05)	-0.36 (-0.46 to -0.26)	-0.16 (-0.33 to 0.01)	-0.40 (-0.50 to -0.30)	-0.20 (-0.37 to -0.04)
	15	-0.24 (-0.34 to -0.14)	0.08 (-0.25 to 0.09)	-0.28 (-0.38 to -0.18)	-0.12 (-0.29 to 0.04)	-0.32 (-0.42 to -0.22)	-0.16 (-0.33 to 0.00)
	16	-0.17 (-0.27 to -0.07)	-0.05 (-0.22 to 0.12)	-0.21 (-0.31 to -0.11)	-0.09 (-0.26 to 0.08)	-0.25 (-0.35 to -0.15)	-0.13 (-0.30 to 0.04)

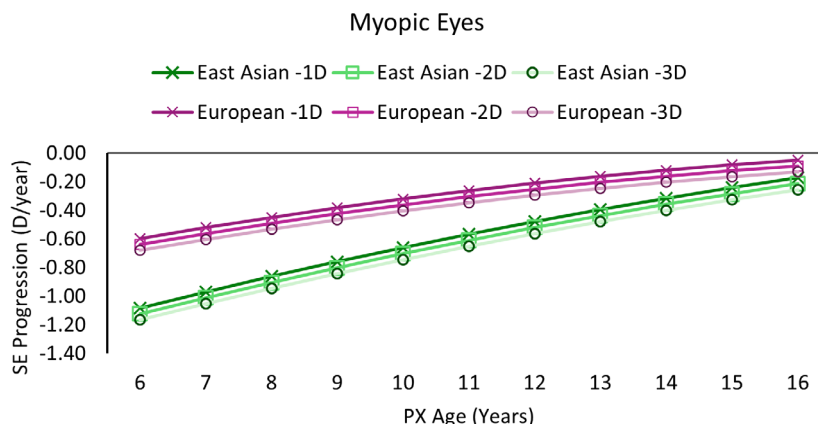


FIGURE 6 Annualised spherical equivalent (SE) progression (D/year) estimated by Generalised Estimating Equation (GEE) model by age at baseline in East Asian and European myopic eyes. PX, participant.

greater than Caucasians in both a younger cohort (-0.28 vs. -0.13 D, respectively) and an older cohort (-0.21 vs. -0.11 D, respectively), suggesting 48%–54% lower progression in Caucasian children.³⁶ In Singapore, where myopia is highly prevalent, myopia progression was shown to be significantly greater in Chinese children (-2.18 D/year) compared with non-Chinese children (-1.71 D/year). In the same cohort, although annual axial elongation did not differ by ethnicity, Chinese children had significantly greater elongation of the vitreous chamber depth compared with non-Chinese children.³⁷ Meta-analysed estimates of axial elongation by Shamp et al. also showed differences by ethnicity across several regions, with a greater growth rate in Asian than Caucasian children and absolute differences decreasing with increasing age,¹⁷ which is similar to the present study's findings, derived directly from patient data. Regional differences have also been shown in growth charts for axial length from populations in China^{9,10} and Europe.^{7,8} These growth charts indicate that the reference values differ between populations in China and Europe. For example, the 50th percentile for a 9-year-old male in China is ~ 0.8 mm higher than their European counterpart and these differences increase with increasing percentiles. Although these growth comparisons are by regions, it is likely that region and ethnicity overlap when comparing myopia data between European and Chinese populations. Accordingly, the present analysis differentiated the sample by regions but ensured there was a more homogeneous ethnic group within a region; thus, the regional differences reported here are reflective of both environmental and ethnic differences. However, the magnitude of ethnic difference is likely to vary by region due to variations in environment and exposure to myopic risk factors, as indicated by Rose et al. who reported that young Asian children in Sydney, Australia had a significantly lower prevalence of myopia and higher exposure to outdoors compared with similarly aged Asian children in Singapore.³⁸

Additionally, this study showed that regional/ethnic differences were greater in the younger ages, a finding that was also observed by Shamp et al.¹⁷ A plausible reason is the greater exposure to education and emphasis on reading prior to starting formal schooling in the East/Southeast Asian region, where myopia is associated with higher educational achievements.^{38–40} These findings highlight the need to identify individuals at risk of progression and commence preventive and therapeutic myopia management at an early age and monitor progression using age-specific tools.

The current investigation also highlights that regional/ethnic differences in axial elongation are present, not only in myopes but also in non-myopes. The definition of non-myopes in the present study excluded incident myopes as it was limited to only a 1-year period of observation. Rozema et al. showed significant differences in axial growth between myopes and emmetropes, where axial growth was higher in incident myopes compared with eyes that remained emmetropic up to 3 years prior to the onset of myopia.⁴¹ The regional/ethnic differences of axial elongation in non-myopes shown here differ from recent analysis by Yii, where differences by ethnicity was not significant in persistent emmetropic eyes.²⁴ The methodology employed by Yii used axial length measurements from cross-sectional studies to compute the annualised emmetropic growth rate in persistent emmetropic eyes, whilst the current study used axial elongation from follow-up data. Moreover, it is possible that our sample of persistent non-myopes included eyes that would become myopic after 1 year, which is more likely in the Asian region. Although the results are not directly comparable due to methodological differences, the present findings highlight the need and the means to identify and monitor high progressing non-myopic children.

Age-specific estimates of axial elongation show a significant age dependence which exponentially reduces with increasing age. Meta-analysed estimates of axial elongation in myopic children by Shamp et al.¹⁷ also showed

an exponential reduction with age. However, their age-specific estimates were higher than the present estimates by an average of 0.06 ± 0.02 mm for both Asian and Non-Asian myopic children. These differences may be attributed to sampling variation, where Shamp et al. relied on clinical trial summary data whereas we used raw data taken primarily from population-based sources, which typically entail a lower proportion of progressing myopes compared with clinical trials. Age-specific axial elongation estimates for myopic and non-myopic East Asian children reported here are concordant with estimates by Naduvilath et al.,¹⁹ with differences within ± 0.02 mm. Differences of this magnitude are within instrument variability,⁴² and expected given the inclusion of data from Vietnam and the estimates being derived from a single model using data from multiple regions.

In terms of age-specific estimates of SE progression, there are few sources with which to compare these estimates. Meta-analysed clinical studies from urban populations estimated myopia progression for a 9.3-year-old myopic Asian to be -0.83 D at the first year of follow-up versus -0.52 D for a European myopic child.¹³ Estimates from the present model were within ± 0.03 D in East Asian children. However, for European children, our estimate was lower by 0.10 D. Slightly lower estimates are expected given that our data source had a higher proportion of population-based studies and used individual patient data rather than summary means.

Using the Brien Holden Vision Institute (BHVI) myopia calculator,⁴³ the age-specific progression estimates for a child starting at -2.0 D are marginally higher than our estimates for myopic East Asian children by 0.05 ± 0.07 D, with larger differences at 14 years of age and beyond. For myopic non-Asian children, the BHVI myopia calculator estimates are lower by 0.07 D at ages 6–7 years and higher by 0.09 D at 14 years and older. These small, clinically insignificant variations could be attributed to the historical data sources used, which were predominantly clinical trial sources, and the higher weights applied to more recent datasets in the present analysis. Interestingly, our age-specific progression for a European eye at -2.0 D is lower than an Asian eye by 50% at 11 years of age. This is similar to differences derived with the BHVI calculator, although our analysis highlights that these differences are age-specific.

The current model estimates suggest that regional/ethnic differences in relative terms were larger in SE progression than in axial elongation. This is also evidenced by the scatter plots that show that at low levels of axial elongation, East Asian eyes demonstrated higher SE progression than European eyes, thus resulting in a greater regional/ethnic difference for SE progression. Liu et al. summarised this relationship using a ratio of SE progression to axial elongation, which ranged for Chinese myopic eyes from 2.06 to 2.59 D/mm.³⁵ Whilst the East Asian myopic children in the present study fell within this range (1.9–2.8 D/mm), the European children were significantly lower, ranging from

1.4 to 2.0 D/mm. Similarly, derived ratios from age-specific estimates of biometry and refraction by Twelker et al.¹⁴ indicated that the ratio was lower in Caucasian children compared with Asian children. They also observed that Asian children were not represented in the extreme ends of the ocular biometry distribution despite highly myopic refractions, suggesting a complex risk factor model that varied by region and ethnicity.

Other than age, sex was the only demographic factor considered for this analysis. The model indicated that non-myopic females had higher axial elongation and myopic females had higher SE progression. The literature on progression by sex is mixed, with female children implicated for higher progression in Singapore³⁷ and Finland⁴⁴ but not in Hong Kong.⁴⁵ The magnitude of the sex difference from our analysis indicated that it was of low clinical significance.

These estimates of axial elongation and myopia progression from children in China, Vietnam, Australia, Sweden and the UK could contribute towards normative data specific to age, sex and region/ethnicity. Historical normative data from untreated myopes derived from Singapore and USA were used to compare axial elongation when using MiSight contact lenses ([misight.com](https://www.misight.com)) by Chamberlain et al.⁴⁶ When children cannot be enrolled into a single vision arm of a myopia control trial, Bullimore et al.²⁵ suggested alternative strategies such as historical normative data derived from meta-analysis rather utilising data from a single study. As previously mentioned, Shamp et al.'s meta-analysis from over 40 clinical trials yielded axial elongation estimates that were higher than the current estimates by $\sim 20\%$, and a similar difference was observed for myopia progression in European children when compared with Donovan's meta-analysis of 20 clinical trials.^{13,17} Both of these meta-analyses used only clinical trial data, which is likely to be biased towards progressing myopes. By contrast, annualised population estimates from Singaporean children aged 7–9 years were comparable to estimates of myopia progression for East Asian children in the present study.³⁷ Whilst the dataset used in the present analysis has the advantage of generalisability to a wider population and narrow confidence bands due to the use of population-based studies, raw data and large sample sizes, a meta-analysis that includes multiple populations and clinic-based studies across several regions would be valuable to develop robust population and age-specific virtual control groups. Age-specific estimates and confidence limits from such a meta-analysis could then be used to establish the efficacy of new treatments. It is acknowledged that environmental factors associated with myopia are changing over time across the world,⁴⁷ and this will require more updated estimates of progression if historical normative data is to be used for product comparisons in clinical trials.

Whilst we have attempted to estimate regional differences in myopia progression, this work is not without limitations. First, the definition of ethnicity is likely to be vague as these were self- or parent-reported ethnicity. Due to this limitation, general terms such as East Asian and

European were used rather than more specific terms, such as Caucasian, Han or Vietnamese. Second, the comparison of the regions could be questioned in terms of the level of urbanisation. The majority of our European data did not represent highly urban areas within Europe. We attempted to reduce this bias by giving greater weight to more recent data (2021) relative to that collected in 2005. However, this method assumes the same rate of urbanisation over 16 years between regions. More data are needed to compare these estimates to other regions, especially non-Asian areas. Third, instruments used for axial length and refraction varied between studies and regions. Most studies used an open-field auto-refractor, whilst two studies used the Topcon auto-refractor, which is not open-field. However, inter-study differences in myopia progression data are likely to be low as all study protocols used cycloplegia. Although a variety of axial biometers were used across studies, all employed partial coherence interferometry-based biometers and the general agreement between instruments is reported.³⁴ Fourth, filters applied to the dataset to identify outliers and children who may have used myopia control may not be sufficiently sensitive. However, these numbers are likely to be small and their impact on the model estimates may be minimal as all study protocols excluded children using myopia control. Finally, using a definition of incident myopia based on 1 year of follow-up may not be sufficiently long to classify persistent non-myopes accurately, which may impact model estimates.

In summary, estimates of axial elongation and myopia progression show that substantial regional/ethnic differences exist. These differences exist in both myopic and non-myopic children and are age-specific, with differences reducing with increasing age. Model-based estimates by age, gender and region/ethnicity may be used to assess myopia treatment efficacy.

AUTHOR CONTRIBUTIONS

Thomas Naduvilath: Conceptualization (lead); data curation (equal); formal analysis (lead); writing – original draft (lead); writing – review and editing (equal). **Xiangui He:** Data curation (equal); funding acquisition (lead); investigation (lead); resources (equal); supervision (equal); writing – review and editing (supporting). **Kathryn Saunders:** Data curation (supporting); funding acquisition (lead); investigation (equal); resources (equal); supervision (equal); writing – original draft (supporting); writing – review and editing (equal). **Pelsin Demir:** Data curation (lead); investigation (equal); writing – review and editing (equal). **Rebecca Leighton:** Data curation (lead); investigation (equal). **Sara McCullough:** Data curation (lead); investigation (equal); resources (equal); writing – review and editing (equal). **Huy Tran:** Data curation (supporting); investigation (equal); resources (equal); supervision (equal); writing – review and editing (supporting). **Thao Ha:** Data curation (lead); investigation (equal). **Antonio Filipe Macedo:** Data curation (supporting); investigation (equal); resources (equal); supervision (equal); writing – review and editing (supporting).

Xu Xun: Funding acquisition (equal); investigation (supporting); resources (equal); supervision (equal). **Padmaja Sankaridurg:** Funding acquisition (equal); project administration (equal); resources (equal); supervision (equal); writing – original draft (supporting); writing – review and editing (equal). **Nina Tahhan:** Conceptualization (supporting); project administration (equal); resources (equal); supervision (equal); writing – original draft (supporting); writing – review and editing (equal).

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CONFLICT OF INTEREST STATEMENT

Thomas Naduvilath: Employee of Brien Holden Vision Institute (BHVI); Nina Tahhan: Consultant at BHVI; Xiangui He and Xun Xu: Employees of Shanghai Eye Disease Prevention and Treatment Centre (SEDPTC). BHVI has a collaboration agreement with SEDPTC. BHVI hosts the BHVI myopia calculator, a free web-based myopia tool, and has commercial interests in myopia management. Kathryn Saunders and Sara McCullough: Employees of Ulster University, hosts PreMO app, a free web-based myopia tool, and receive industry funding for myopia research. Padmaja Sankaridurg: Former employee of BHVI, a current employee of ZEISS VisionCare, a commercial entity involved in myopia management. No other conflict of interest relevant to this manuscript exists for any of the authors.

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